The location of the most isolated point in a polyhedral sample

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Abstract

Let A be a compact convex polyhedron in $\mathbb{R}^3$ with nonempty interior, and sample independent uniform points in A. We prove that the point with the largest nearest-neighbour distance converges in distribution to normalized surface area on ∂A when every interior dihedral angle exceeds $\pi/2$. When the smallest angle is less than $\pi/2$, its limit is normalized length on the union of all edges with that angle. The proof computes spatial empty-ball intensities and joint exceedance probabilities for the fixed-size sample. The critical angle is excluded.

1 The location question

In a geometric graph with connection radius r, a sample point is isolated precisely when its nearest-neighbour distance exceeds r. The point whose isolation persists longest therefore has the largest such distance. The boundary affects both the size of this distance and the location of the point attaining it: an empty ball costs less near a face or an edge, but fewer sample points are available there.

Let $A \subset \mathbb{R}^3$ be a fixed compact convex polyhedron with nonempty interior, and write $V = |A|$ for its volume. Let $X_1, X_2, \dots$ be independent uniform points in A. For $n \geq 2$ put

$$R_{n,i} = \min_{\substack{1 \leq j \leq n \\ j \neq i}} \|X_i - X_j\|, \qquad L_n = \max_{1 \leq i \leq n} R_{n,i}, \qquad I_n = \min\{i : R_{n,i} = L_n\}, \qquad Z_n = X_{I_n}.$$

For an edge e, let $\alpha_e \in (0, \pi)$ denote its interior dihedral angle, and set

$$\alpha = \min_e \alpha_e, \qquad E_* = \bigcup_{\alpha_e = \alpha} e, \qquad S = \mathcal{H}^2(\partial A), \qquad L = \mathcal{H}^1(E_*).$$

Here $\mathcal{H}^j$ has the usual Euclidean normalization, so these are surface area and edge length. Both S and L are positive and finite.

Penrose, Yang and Higgs proved the leading strong law

$$\frac{nL_n^3}{V \log n} \longrightarrow \max\left\{\frac{1}{\pi}, \frac{1}{2\alpha}\right\} \quad \text{almost surely}$$

and proposed the corresponding spatial limits [1, Introduction, p. 1724; Corollary 2.3]. Their strong law identifies the competition between faces and sharp edges, but does not determine the spatial distribution of the maximizer. Later fluctuation results of Penrose and Yang treat smooth domains and planar polygons [2, arXiv v3, Theorem 1.1]; general three-dimensional polyhedra are outside that theorem's hypotheses. We prove the two proposed spatial limits.

Theorem 1. *For every compact convex polyhedron $A \subset \mathbb{R}^3$ with nonempty interior, the point Z_n defined above satisfies*

$$Z_n \xrightarrow{d} \begin{cases} \mathcal{H}^2|_{\partial A}/S, & \alpha > \pi/2, \\ \mathcal{H}^1|_{E_*}/L, & \alpha < \pi/2. \end{cases}$$

The statement includes all minimum-angle edges and the specified smallest-index rule for ties. The case $\alpha = \pi/2$ is not considered.

We first compute the intensity of points with an empty neighbourhood at an extreme radius. Its spatial weight is constant per unit area on a facet, and constant per unit length on edges with the same angle. To identify the maximizer, we also need joint probabilities. A convex-geometric volume increment makes nearby simultaneous extreme events negligible; factorial moments then give joint no-exceedance limits. Comparing maxima in a set and its complement proves Theorem 1.

All estimates below concern this fixed A. Constants may also depend on a fixed finite set of thresholds or a fixed moment order. Balls are Euclidean and closed. We use the elementary product-integration and convergence results in [3, Sections 1.5, 1.7 and 3.10], and the linear volume transformation formula in [4, Section 2.8].

2 Local polyhedral geometry

For $x \in A$ and $r > 0$, write

$$v_r(x) = |A \cap B(x, r)|.$$

Two estimates will control the vertices and the dependence between nearby empty-ball events.

Lemma 2. *There is $b > 0$ such that $v_r(x) \geq br^3$ for every $x \in A$ and all sufficiently small r. Moreover, for every fixed $M \geq 1$ there are $c > 0$ and $r_0 > 0$ such that*

$$|A \cap (B(x, r) \cup B(y, r))| \geq \min\{v_r(x), v_r(y)\} + cr^3 \tag{1}$$

whenever $0 < r < r_0$ and $r \leq \|x - y\| \leq Mr$.

Proof. Choose $z \in A$, $\rho > 0$ and $D \geq \sup_{x \in A} \|z - x\| + \rho$ with $B(z, \rho) \subset A$. Convexity gives

$$B(x + u(z - x), u\rho) \subset A \qquad (0 \leq u \leq 1). \tag{2}$$

For $u = r/(2D)$ this ball is contained in $B(x, r)$, proving the first assertion with $b = (4\pi/3)(\rho/(2D))^3$.

For the second assertion let $d = \|x - y\|$, $q = (x + y)/2$ and $u = (y - x)/d$. If $\|q - z\| < \rho/2$, then for r sufficiently small both radius-r balls lie inside $B(z, \rho)$. The ball $B(y + (r/2)u, r/8)$ lies in $B(y, r)$ and outside $B(x, r)$.

Suppose instead that $\|q - z\| \geq \rho/2$. Choose $w = y$ if $u \cdot (z - q) \geq 0$, and $w = x$ otherwise. Let w' be the other point, $u_w = (w - w')/d$, $\theta = r/(4D)$, and $p = w + \theta(z - w)$. Then $u_w \cdot (z - q) \geq 0$, $B(p, \theta\rho) \subset A$, and $\|p - w\| \leq r/4$. Also

$$\begin{aligned} \|p - w'\|^2 &= \|(1 - \theta/2)du_w + \theta(z - q)\|^2 \\ &\geq (1 - \theta)d^2 + \theta^2\rho^2/4 \\ &\geq r^2 - \frac{M^2 r^3}{4D} + \frac{\rho^2 r^2}{64D^2}. \end{aligned}$$

For small r, this is at least $(1+\rho^2/(128D^2))r^2$. Hence $\|p-w'\| \geq r+\eta_0 r$ for some $\eta_0 > 0$ depending on A. Choose $0 < \eta < \min\{\eta_0/2, \rho/(8D), 1/8\}$. The ball $B(p, \eta r)$ lies in $A \cap B(w, r)$ and outside $B(w', r)$.

In either case the union contains one of the clipped balls and an exclusive ball of volume at least a fixed positive multiple of r^3. Subtracting the smaller individual volume proves (1). □

We will also use (1) for radii between r_- and r_+ with $r_+/r_- \to 1$. The volume change in increasing a radius from r_- to r_+ is at most $(4\pi/3)(r_+^3 - r_-^3) = o(r_-^3)$. Consequently, if $\max\{r_i, r_j\} \leq \|x-y\| \leq M r_+$, then for large n

$$|A \cap (B(x, r_i) \cup B(y, r_j))| \geq \min\{v_{r_i}(x), v_{r_j}(y)\} + c' r_-^3. \tag{3}$$

Here Lemma 2 is applied at r_-, with a fixed enlargement of M, and half its increment absorbs the shell-volume error.

We next partition A into regions where a small ball sees a full space, a half-space or a wedge. Let $\mathcal{F}$, $\mathcal{E}$ and $\mathcal{V}$ be the facets, edges and vertices of A. For $f \in \mathcal{F}$, let d_f be the affine inward signed distance to its supporting plane. Thus $A = \{x : d_f(x) \geq 0 \text{ for all } f \in \mathcal{F}\}$, with irredundant facet inequalities.

Lemma 3. *There are $C > 0$ and $r_0 > 0$ such that, for $0 < r < r_0$, outside $\bigcup_{v \in \mathcal{V}} B(v, Cr)$ the set*

$$J_r(x) = \{f \in \mathcal{F} : d_f(x) < r\}$$

is empty, a singleton, or the two facets incident to an edge. The singleton region for f lies between the coordinate domains $f'_r \times [0, r]$ and $f \times [0, r]$, with $\mathcal{H}^2(f \setminus f'_r) = O(r)$. The two-facet region for e lies between $e'_r \times [0, r]^2$ and $e \times [0, r]^2$, with $\mathcal{H}^1(e \setminus e'_r) = O(r)$. In the edge coordinates (y, h_1, h_2), where $y \in e$ is orthogonal projection and h_i is inward face distance,

$$dx = \frac{d\mathcal{H}^1(y)\, dh_1\, dh_2}{\sin \alpha_e}. \tag{4}$$

All domain inclusions are understood up to null boundaries.

Proof. A point of a positive-dimensional face is a convex combination of its vertices: a maximal line segment through a point in its minimal face has endpoints on proper subfaces, and induction on face dimension gives the assertion. The vertices are finite because the facet planes are finite.

For a nonempty subset $J \subset \mathcal{F}$, let $F_J = A \cap \{d_f = 0 : f \in J\}$ and $D_J = \sum_{f \in J} d_f$. If $F_J \neq \varnothing$, D_J vanishes on its vertices and is strictly positive at every other vertex. Let $\delta_J > 0$ be the minimum of these positive values. In a vertex convex combination for x, the total weight β on vertices outside F_J is at most $D_J(x)/\delta_J$. When this is less than one, normalizing the remaining weights gives a point of F_J within distance $\operatorname{diam}(A)\beta$ of x. Thus

$$\operatorname{dist}(x, F_J) \leq C_J D_J(x) \qquad \text{when } D_J(x) \text{ is small.}$$

If F_J is empty, D_J has a positive minimum on compact A. There are finitely many J, so an active subset $J_r(x)$ has nonempty common face and puts x within $O(r)$ of that face.

Outside sufficiently large Cr vertex balls, the common face has positive dimension. A facet is specified by a single irredundant inequality. An edge has exactly two incident maximal facets, since a transverse slice at a relative-interior edge point is a planar wedge with two boundary rays. Three distinct maximal facets cannot contain an edge. This proves the asserted alternatives and includes nonsimple vertices.

For a singleton f, write $x = y + hu_f$ with u_f its inward unit normal and $0 \le h < r$. Every other slack is at least r, so $d_g(y) \ge d_g(x) - h \ge 0$ and $y \in f$. For a lower domain take those $y \in f$ with $d_g(y) \ge 2r$ for all $g \ne f$ and $\operatorname{dist}(y, \mathcal{V}) \ge (C+1)r$. These conditions ensure that only f is active and the vertex balls are avoided. Each nonconstant affine restriction $d_g|_f$ removes a strip of width $O(r)$ from a bounded polygon; constant restrictions are strictly positive. Vertex disks remove $O(r^2)$ area. The total area lost is $O(r)$.

For an edge e, the map from transverse orthonormal coordinates to the two inward face distances has determinant of absolute value $\sin \alpha_e$, because the inward normals have angle $\pi - \alpha_e$. This proves (4). If $0 \le h_i < r$, the transverse displacement is at most $C_e r$. The face-distance bound also gives $\operatorname{dist}(x, e) \le C'_e r$. A projection beyond an endpoint would therefore put x in an $O(r)$ vertex ball; enlarge C to exclude it. For a lower domain trim $O(r)$ length from the ends of e, so all nonincident slacks at y exceed $(C_e + 1)r$ and $\operatorname{dist}(y, \mathcal{V}) > (C + C_e)r$. Each such slack is affine and positive in the relative interior, so the required trimming has size $O(r)$. The transverse displacement preserves slack at least r and distance greater than Cr from the vertices. This proves the lower inclusion. All constants may be chosen simultaneously over the finitely many faces. □

3 Spatial empty-ball intensities

Put $\lambda_n = n/V$. The conditional probability that a sample point at x has no other point within distance r is $(1 - v_r(x)/V)^{n-1}$. At the radii used below it is uniformly asymptotic to $e^{-\lambda_n v_r(x)}$. We begin with the latter expression.

For fixed $s \in \mathbb{R}$, define $r_n(s) > 0$ for sufficiently large n by $t_n(s) = \lambda_n r_n(s)^3$, where

$$t_n(s) = \begin{cases} \pi^{-1}(\log n - \log\log n) + (3/(2\pi))s, & \alpha > \pi/2, \\ (2\alpha)^{-1}(\log n - 4\log\log n) + (3/(2\alpha))s, & \alpha < \pi/2. \end{cases} \tag{5}$$

Set

$$(\nu, K) = \begin{cases} \left(\mathcal{H}^2|_{\partial A}/S,\ S/(\pi^{1/3}V^{2/3})\right), & \alpha > \pi/2, \\ \left(\mathcal{H}^1|_{E_*}/L,\ 4(2\alpha)^{4/3}L/(\pi^2 \sin\alpha\, V^{1/3})\right), & \alpha < \pi/2. \end{cases} \tag{6}$$

Proposition 4. *For every fixed $s \in \mathbb{R}$, the finite measures on A*

$$\mu_{n,s}(dx) = \lambda_n e^{-\lambda_n v_{r_n(s)}(x)}\, dx$$

converge weakly to $Ke^{-s}\nu$. The same limit holds on every ν-continuity Borel set $B \subset A$ for the expected number of indices with $X_i \in B$ and $R_{n,i} > r_n(s)$.

Proof. Let $r = r_n(s)$, $t = \lambda_n r^3$, and let ψ be continuous on A. In a half-space the unit-ball volume at inward center distance $u \in [0,1]$ is

$$g_F(u) = \frac{2\pi}{3} + \pi u - \frac{\pi}{3}u^3.$$

This follows by integrating the disk sections from height $-u$ to 1. Since $g_F(u) - g_F(0) \ge (2\pi/3)u$ and $g'_F(0) = \pi$, rescaling $u = w/t$ gives

$$\int_0^1 e^{-t g_F(u)}\, du \sim \frac{e^{-2\pi t/3}}{\pi t}. \tag{7}$$

The rescaled integrand is dominated on $\mathbb{R}_+$ by $e^{-(2\pi/3)w}$ and tends to $e^{-\pi w}$.

For an edge of angle $\beta \in (0,\pi)$ let a_1, a_2 be its inward unit normals and define

$$g_\beta(u_1,u_2) = \left|\{z \in B(0,1) : a_i \cdot z \geq -u_i,\ i=1,2\}\right|.$$

At $(0,0)$ the wedge occupies fraction $\beta/(2\pi)$ of the ball, so $g_\beta(0,0) = 2\beta/3$. Relaxing one inequality adds a slab whose cross-sectional area tends to half a unit disk, $\pi/2$. The remaining doubly negative region has third-coordinate length at most 2 and face-distance area at most $u_1u_2/\sin\beta$. It follows that

$$g_\beta(u_1,u_2) = \frac{2\beta}{3} + \frac{\pi}{2}(u_1+u_2) + o(u_1+u_2). \tag{8}$$

Slice-area continuity follows from dominated convergence, since the limiting straight-line boundary has area zero. The function g_β is continuous and coordinatewise nondecreasing, and an increase of either coordinate from zero adds positive volume. Compactness of $[0,1]^2$ and (8) give

$$g_\beta(u_1,u_2) - \frac{2\beta}{3} \geq c_\beta(u_1+u_2) \qquad (0 \leq u_i \leq 1)$$

for some $c_\beta > 0$. Rescaling both coordinates by t and using this integrable exponential bound yields

$$\int_{[0,1]^2} e^{-tg_\beta(u_1,u_2)}\,du_1du_2 \sim \frac{4e^{-2\beta t/3}}{\pi^2 t^2}. \tag{9}$$

Apply Lemma 3. Replacing $\psi(x)$ by its facet or edge projection changes it by at most its modulus of continuity at C_Ar. The upper and lower domains differ by $O(r)$ area or length, multiplied by the same positive transverse integral. Thus the contribution of a facet f is

$$\frac{\lambda_n r e^{-2\pi t/3}}{\pi t}\left(\int_f \psi\,d\mathcal{H}^2 + o(1)\right)(1+o(1)), \tag{10}$$

and that of an edge e of angle β is

$$\frac{4\lambda_n r^2 e^{-2\beta t/3}}{\pi^2 \sin\beta\, t^2}\left(\int_e \psi\,d\mathcal{H}^1 + o(1)\right)(1+o(1)). \tag{11}$$

These estimates also hold for signed ψ by the same absolute error bound. The full-ball region contributes at most $\|\psi\|_\infty n e^{-4\pi t/3}$ in absolute value. The vertex region has volume $O(r^3)$, and Lemma 2 bounds its contribution by $C_A\|\psi\|_\infty t e^{-bt}$.

In the face regime, $t \sim \pi^{-1}\log n$. The prefactor in (10) tends to $\pi^{-1/3}V^{-2/3}e^{-s}$. The full-ball term is $O_s(n^{-1/3}(\log n)^{4/3})$, and the vertex term tends to zero. For an edge of angle β, its prefactor in (11) is a constant times

$$n^{1/3-2\beta/(3\pi)}(\log n)^{-4/3+2\beta/(3\pi)}.$$

Every $\beta \geq \alpha > \pi/2$, so these terms vanish. Summing the facet integrals gives $Ke^{-s}\int\psi\,d\nu$ with the first row of (6).

In the edge regime, $t \sim (2\alpha)^{-1}\log n$. A minimum-angle edge has prefactor tending to $4(2\alpha)^{4/3}e^{-s}/(\pi^2\sin\alpha\, V^{1/3})$. For another edge the power of n is $1/3-\beta/(3\alpha) < 0$. For facets it is $2/3-\pi/(3\alpha) < 0$, and for the full-ball region it is $1-2\pi/(3\alpha) < 0$. All accompanying logarithmic powers are fixed. The vertex term again vanishes. Only edges of angle α remain, and their identical prefactors give the second row of (6). This proves the finite-measure weak limit.

Finally, $v_r(x) \leq (4\pi/3)r^3 = O(\log n/n)$ uniformly. For fixed k,

$$(1-v_r(x)/V)^{n-k} = e^{-\lambda_n v_r(x)}(1+o(1)), \tag{12}$$

uniformly, since the logarithm error is $O_k((\log n)^2/n)$. Taking $k=1$ identifies the actual first moments. The total mass of $\mu_{n,s}$ tends to $Ke^{-s} > 0$; normalizing it gives probability measures converging weakly to ν. The continuity-set characterization of weak convergence, followed by multiplication by the total mass, proves the asserted limits on B. □

4 Joint no-exceedance limits

We now keep the spatial categories while calculating the joint events in the original sample of exactly n points.

Proposition 5. *Let $B_1, \dots, B_m$ be pairwise disjoint ν-continuity Borel subsets of A, and let $s_1, \dots, s_m \in \mathbb{R}$ be fixed. If W_n counts the indices i for which $X_i \in B_j$ and $R_{n,i} > r_n(s_j)$ for some j, then*

$$\mathbb{P}(W_n = 0) \longrightarrow \exp\left(-K \sum_{j=1}^{m} \nu(B_j) e^{-s_j}\right). \tag{13}$$

Proof. Fix $k \geq 1$ and take $n > k$. Expand the falling factorial moment $\mathbb{E}[(W_n)_k]$ over the spatial categories of k ordered selected indices. For one such assignment let r_i be the radius of the category of x_i. Conditional on these locations, the joint exceedance event is impossible unless

$$\|x_i - x_l\| > \max\{r_i, r_l\} \qquad (i \neq l). \tag{14}$$

When this holds, the remaining $n-k$ points must avoid the ball union, so the corresponding factorial integral is

$$\frac{(n)_k}{V^k} \int_{B_{j_1} \times \cdots \times B_{j_k}} \mathbf{1}_{(14)} \left(1 - \frac{|A \cap \bigcup_{i=1}^{k} B(x_i, r_i)|}{V}\right)^{n-k} dx_1 \cdots dx_k. \tag{15}$$

Here $(w)_k = w(w-1)\cdots(w-k+1)$ and $(w)_0 = 1$ for nonnegative integers w. The indicator in (15) enforces (14). Let r_- and r_+ be the smallest and largest of the finitely many $r_n(s_j)$. Their ratio tends to one, and $\lambda_n r_\pm^3$ is asymptotic to a positive multiple of $\log n$.

If all pair distances exceed $2r_+$, the balls are disjoint. Their union volume is $O_k(\log n/n)$, so the same logarithm estimate as in (12) changes the probability in (15), uniformly, to

$$(1 + o(1)) \prod_{i=1}^{k} e^{-\lambda_n v_{r_i}(x_i)}.$$

The unrestricted product integral has limit $\prod_i K e^{-s_{j_i}} \nu(B_{j_i})$, by Proposition 4 and $(n)_k/n^k \to 1$. The part removed from this product by a pair at distance at most $2r_+$ is $o(1)$. Indeed one point has a scaled integral $O(1)$, its partner is integrated over volume $O(r_+^3)$ and has exponential weight at most $e^{-b\lambda_n r_-^3}$, and all other scaled integrals are $O(1)$. The resulting bound is

$$C_k \lambda_n r_+^3 e^{-b\lambda_n r_-^3} = o(1). \tag{16}$$

It remains to bound compatible configurations with a close pair in the actual integral (15). Join two of the k centers when their distance is at most $2r_+$. Balls in distinct connected components are disjoint. In each component choose a root with the smallest individual volume $v_{r_i}(x_i)$; there are only finitely many possible root choices. A component with at least two vertices has an edge satisfying (14). By (3), its union volume is at least the root volume plus cr_-^3: the smaller volume of this pair is at least the minimum over the component.

Every center of a component lies within $2(k-1)r_+$ of its root. If it has q vertices, integration of its other $q-1$ coordinates costs at most $C_k r_+^{3(q-1)}$. Put $\lambda_n' = (n-k)/V$, and use $(1-u)^{n-k} \leq e^{-(n-k)u}$ in the original conditional probability. Each root satisfies

$$\lambda_n \int_A e^{-\lambda_n' v_{r_i}(x)}\, dx = O(1),$$

because replacing λ_n by λ'_n changes its integrand by $1+o(1)$ uniformly. If there are ℓ components and at least one is nontrivial, the scaled contribution is therefore at most

$$C_k(\lambda_n r_+^3)^{k-\ell} e^{-c\lambda'_n r_-^3} = o(1). \tag{17}$$

For fixed k, the number of component partitions and root choices is finite. The exponential in (17) is a negative power of n, while its prefactor is a fixed power of $\log n$. Category and separation constraints can be dropped when bounding the nonroot volumes and root integrals. Summing all assignments gives, with $a = K\sum_j \nu(B_j)e^{-s_j}$,

$$\mathbb{E}[(W_n)_k] \longrightarrow a^k \qquad \text{for every fixed } k \geq 0.$$

For an integer $w \geq 1$, Pascal's identity gives

$$\sum_{k=0}^{h} (-1)^k \binom{w}{k} = (-1)^h \binom{w-1}{h},$$

with terms above w taken as zero. For $w = 0$ the sum is one. Its odd truncations bound $\mathbf{1}_{\{w=0\}}$ from below and its even truncations from above. Hence, for every fixed $h \geq 0$,

$$\sum_{k=0}^{2h+1} \frac{(-1)^k \mathbb{E}[(W_n)_k]}{k!} \leq \mathbb{P}(W_n = 0) \leq \sum_{k=0}^{2h} \frac{(-1)^k \mathbb{E}[(W_n)_k]}{k!}.$$

First send n to infinity, and then h to infinity. Both exponential series converge to e^{-a}, proving (13). □

5 The maximizing point

Proof of Theorem 1. The increasing score whose level s corresponds to $r_n(s)$ is

$$T_n(r) = \begin{cases} (2\pi/3)\lambda_n r^3 - (2/3)\log n + (2/3)\log\log n, & \alpha > \pi/2, \\ (2\alpha/3)\lambda_n r^3 - (1/3)\log n + (4/3)\log\log n, & \alpha < \pi/2. \end{cases}$$

Let B be a ν-continuity Borel subset of A, and put $q = \nu(B)$. Suppose first that $0 < q < 1$. Let $M_{n,B}$ be the maximum of $T_n(R_{n,i})$ over points in B, with empty maximum $-\infty$, and define $M_{n,A\setminus B}$ in the same way. Proposition 5 at thresholds s, t gives

$$\mathbb{P}(M_{n,B} \leq s,\ M_{n,A\setminus B} \leq t) \longrightarrow e^{-Kqe^{-s}} e^{-K(1-q)e^{-t}}. \tag{18}$$

An empty maximum in B is at most $-H$ for every fixed $H > 0$; its limsup probability is consequently at most e^{-Kqe^H}, which tends to zero as $H \to \infty$. The same holds for the complement. Replacing empty maxima by zero changes any joint CDF or comparison probability by $o(1)$. Thus the multivariate CDF criterion applies to real-valued maxima and yields a pair of independent continuous limiting variables G_B, G_C with the CDF in (18).

The variables e^{-G_B} and e^{-G_C} are independent exponentials with rates Kq and $K(1-q)$. Therefore

$$\mathbb{P}(G_B > G_C) = \int_0^\infty Kq\, e^{-Kqu} e^{-K(1-q)u}\, du = q, \qquad \mathbb{P}(G_B = G_C) = 0.$$

The strict and nonstrict comparison sets in $\mathbb{R}^2$ have this zero-mass diagonal boundary. Since the score is increasing,

$$\mathbb{P}(M_{n,B} > M_{n,A\setminus B}) \le \mathbb{P}(Z_n \in B) \le \mathbb{P}(M_{n,B} \ge M_{n,A\setminus B}).$$

Both bounds tend to q. This retains the specified tie rule, including the possibility that mutual nearest neighbours have the same distance.

If $q = 0$, for every fixed s the event $Z_n \in B$ implies either an exceedance in B above $r_n(s)$, or $L_n \le r_n(s)$. The first probability is bounded by its expected count, which tends to zero. The second tends to $e^{-Ke^{-s}}$ by Proposition 5 with $B_1 = A$. Sending $s \to -\infty$ after $n \to \infty$ gives $\mathbb{P}(Z_n \in B) \to 0$. For $q = 1$ apply this argument to the complement. Thus the probabilities converge to $\nu(B)$ for every continuity set, proving weak convergence on A and hence on $\mathbb{R}^3$. □

AI-use statement. The mathematical argument, source checks and manuscript were prepared with OpenAI Codex. Independent mathematical review is requested.

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